RESHA NAIK

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TECHNICAL SKILLS

Microcontrollers: ESP32, Heltec WiFi LoRa 32 V3, Arduino Uno, Arduino Nano
Sensors & Modules: MQ2, MQ7, ZE16B, PT100/MAX31865, ADXL345, MPU9250, BMP388, MP3 Serial Module, GPS
Communication Protocols: LoRa (SX1278), SPI, UART, I2C, MQTT, WiFi
PCB Design: KiCad – Schematic Capture, PCB Layout, Flexible & Modular Form Factors
Embedded Programming: Embedded C, C++, Arduino IDE
Hardware Debugging: Breadboard prototyping, serial monitoring, multimeter testing
Other: Python, Git, HTML/CSS, JavaScript, Figma

EXPERIENCE

AICTE IDEA Lab Intern – IoT & Embedded Systems Jun 2025 – Aug 2025
SAKEC *Mumbai, India*

- Completed intensive hands-on training in IoT system design – programmed Arduino Uno, Arduino Nano, and ESP32 microcontrollers, implemented MQTT and GPS-based communication, and built and soldered PCB circuits from scratch.
- Applied training to develop a full engineering project (FAiMN) alongside a 4-member team under faculty mentorship, completing prototype development, documentation, and a successful patent filing within the internship period.

UX Designer Intern Feb 2026 – Apr 2026
Godamwale *Mumbai, India*

- Mapped end-to-end warehouse operator workflows to identify 5+ inefficiencies; delivered structured user flows and prototypes across 8+ screens, reviewed directly with founders.

PROJECTS

FAiMN: Firefighters' Assistant IoT based Monitoring Network | *Heltec Esp32, LoRa, KiCad, Embedded C*

- **Patent No. 20262101825** – Patent Journal of India No. 14/2026, Apr 3, 2026 | **Winner: SAKEC Creathon (2025), TechExpo Pillai HOC (2026), Avishkar FCRIT (2026)**
- Integrated 10+ sensors (MQ2, MQ7, ZE16B, PT100, ADXL345, MPU9250, BMP388) onto an ESP32 wearable platform via SPI/UART/I2C, enabling real-time monitoring of gas, temperature, motion, and altitude – validated across 6–7 floors in field testing.
- Iterated 3 hardware prototypes in KiCad (neckband PCB form factor), validating each on breadboard before fabrication; final prototype supports 2 simultaneous users with zero measurable transmission latency over LoRa.

Assistive Suction Glove | *MOSFET Circuit, 12V DC Pump, Pneumatics*

- Built a 12V pneumatic assistive glove for a physiotherapy tremor-patient research project – engineered a MOSFET-based pump control circuit (12V Li-ion, push-button actuation) and handled all mechanical assembly: tubing, 3 suction cups, and glove integration.
- Designed to enable object pickup without grip force; usability validation planned with 20+ participants.

Blood Bond | *HTML, CSS, JavaScript* | *Deployed*

- Built and deployed a responsive web platform connecting human and pet blood donors with recipients, featuring secure authentication and accessible UI.

EDUCATION

Shah & Anchor Kutchhi Engineering College Nov 2022 – Present
*B.Tech in Electronics & Computer Science | CGPA: **9.17/10** (up to Semester 7)*
Vidya Niketan Junior College Mar 2022
HSC – Science (Grade 11 & 12)
Trinity International School May 2020
IGCSE (Grade 10)

ACHIEVEMENTS & CERTIFICATIONS

- **1st Place** – SAKEC Creathon, Project Competition (2025) | **1st Place** – TechExpo, Pillai HOC (2026) | **2nd Place** – Avishkar, FCRIT (2026)
- Participant – Smart India Hackathon and national ideathons | Creativity Coordinator – Student Council | Chairperson – Kalavrunda Art Club

CERTIFICATIONS

- HTML – Infosys Springboard | Python – Coddy Team | Networking & Cybersecurity – Cisco | WordPress – Coursera